

Exercise 3.1 (140 pts)

Answer Summary

- (i) 3
 - (ii) $\frac{2}{3}$
 - (iii) 3
 - (iv) Does not exist
 - (v) $-\frac{3}{4}$
 - (vi) 15
 - (vii) $\frac{1}{2}$
 - (viii) 0
 - (ix) 1
 - (x) $-\sqrt{2}$
 - (xi) $\frac{1}{3}$
 - (xii) ~~$-\infty$~~ DNE !
 - (xiii) e
 - (xiv) $e^{-1/2}$ or $\frac{1}{\sqrt{e}}$
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(i) (10 pts)

We use the fundamental trigonometric limit $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 3x}{x} &= \lim_{x \rightarrow 0} \frac{\sin 3x}{x} \cdot \frac{3}{3} \\ &= \lim_{x \rightarrow 0} 3 \cdot \frac{\sin 3x}{3x} \\ &= 3 \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}\end{aligned}$$

Let $u = 3x$. As $x \rightarrow 0$, $u \rightarrow 0$.

$$3 \cdot \lim_{u \rightarrow 0} \frac{\sin u}{u} = 3 \cdot 1 = \boxed{3}$$

(ii) (10 pts)

We divide the numerator and the denominator by x .

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 6x} &= \lim_{x \rightarrow 0} \frac{\frac{\sin 4x}{x}}{\frac{\sin 6x}{x}} \\&= \frac{\lim_{x \rightarrow 0} \frac{\sin 4x}{x}}{\lim_{x \rightarrow 0} \frac{\sin 6x}{x}} \\&= \frac{\lim_{x \rightarrow 0} 4 \cdot \frac{\sin 4x}{4x}}{\lim_{x \rightarrow 0} 6 \cdot \frac{\sin 6x}{6x}} \\&= \frac{4 \cdot 1}{6 \cdot 1} = \frac{4}{6} = \boxed{\frac{2}{3}}\end{aligned}$$

(iii) (10 pts)

First, we rewrite $\tan(6t)$ as $\frac{\sin(6t)}{\cos(6t)}$.

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\tan 6t}{\sin 2t} &= \lim_{t \rightarrow 0} \frac{\sin 6t}{\cos 6t \cdot \sin 2t} \\&= \lim_{t \rightarrow 0} \left(\frac{\sin 6t}{6t} \cdot \frac{2t}{\sin 2t} \cdot \frac{6t}{2t} \cdot \frac{1}{\cos 6t} \right) \\&= \left(\lim_{t \rightarrow 0} \frac{\sin 6t}{6t} \right) \cdot \left(\lim_{t \rightarrow 0} \frac{2t}{\sin 2t} \right) \cdot \left(\lim_{t \rightarrow 0} \frac{6}{2} \right) \cdot \left(\lim_{t \rightarrow 0} \frac{1}{\cos 6t} \right) \\&= (1) \cdot (1) \cdot (3) \cdot \left(\frac{1}{\cos 0} \right) \\&= 1 \cdot 1 \cdot 3 \cdot \frac{1}{1} = \boxed{3}\end{aligned}$$

(iv) (10 pts)

We analyze the one-sided limits. The expression is $\frac{\cos \theta}{\sin \theta} = \cot \theta$. As $\theta \rightarrow 0^+$, $\cos \theta \rightarrow 1$ and $\sin \theta \rightarrow 0^+$.

$$\lim_{\theta \rightarrow 0^+} \frac{\cos \theta}{\sin \theta} = \frac{1}{0^+} = +\infty$$

As $\theta \rightarrow 0^-$, $\cos \theta \rightarrow 1$ and $\sin \theta \rightarrow 0^-$.

$$\lim_{\theta \rightarrow 0^-} \frac{\cos \theta}{\sin \theta} = \frac{1}{0^-} = -\infty$$

Since the left-sided and right-sided limits are not equal, the limit does not exist.

(v) (10 pts)

We factor the denominator and separate the limit.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 3x}{5x^3 - 4x} &= \lim_{x \rightarrow 0} \frac{\sin 3x}{x(5x^2 - 4)} \\&= \left(\lim_{x \rightarrow 0} \frac{\sin 3x}{x} \right) \cdot \left(\lim_{x \rightarrow 0} \frac{1}{5x^2 - 4} \right) \\&= \left(\lim_{x \rightarrow 0} 3 \cdot \frac{\sin 3x}{3x} \right) \cdot \left(\frac{1}{5(0)^2 - 4} \right) \\&= (3 \cdot 1) \cdot \left(\frac{1}{-4} \right) = \boxed{-\frac{3}{4}}\end{aligned}$$

(vi) (10 pts)

We can rewrite the expression to use the fundamental trigonometric limit.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{x^2} &= \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} \cdot \frac{\sin 5x}{x} \right) \\&= \left(\lim_{x \rightarrow 0} 3 \frac{\sin 3x}{3x} \right) \cdot \left(\lim_{x \rightarrow 0} 5 \frac{\sin 5x}{5x} \right) \\&= (3 \cdot 1) \cdot (5 \cdot 1) = \boxed{15}\end{aligned}$$

(vii) (10 pts)

Divide the numerator and denominator by θ .

$$\begin{aligned}\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta + \tan \theta} &= \lim_{\theta \rightarrow 0} \frac{\frac{\sin \theta}{\theta}}{1 + \frac{\tan \theta}{\theta}} \\&= \frac{1}{1 + \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right) \left(\lim_{\theta \rightarrow 0} \frac{1}{\cos \theta} \right)} \\&= \frac{1}{1 + (1) \cdot (1)} = \boxed{\frac{1}{2}}\end{aligned}$$

(viii) (10 pts)

We manipulate the expression to use the standard limit.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x} &= \lim_{x \rightarrow 0} \left(\frac{\sin(x^2)}{x^2} \cdot x \right) \\&= \left(\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2} \right) \cdot \left(\lim_{x \rightarrow 0} x \right)\end{aligned}$$

Let $u = x^2$. As $x \rightarrow 0$, $u \rightarrow 0$.

$$\left(\lim_{u \rightarrow 0} \frac{\sin u}{u} \right) \cdot \left(\lim_{x \rightarrow 0} x \right) = (1) \cdot (0) = \boxed{0}$$

(ix) (10 pts)

Divide the numerator and denominator by x .

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{2x}{x + \sin x} &= \lim_{x \rightarrow 0} \frac{2}{1 + \frac{\sin x}{x}} \\ &= \frac{2}{1 + 1} = \boxed{1}\end{aligned}$$

(x) (10 pts)

We rewrite $\tan x$ and simplify.

$$\begin{aligned}\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{\sin x - \cos x} &= \lim_{x \rightarrow \pi/4} \frac{1 - \frac{\sin x}{\cos x}}{\sin x - \cos x} \\ &= \lim_{x \rightarrow \pi/4} \frac{\frac{\cos x - \sin x}{\cos x}}{\sin x - \cos x} \\ &= \lim_{x \rightarrow \pi/4} \frac{-(\sin x - \cos x)}{\cos x(\sin x - \cos x)} \\ &= \lim_{x \rightarrow \pi/4} \frac{-1}{\cos x} = \frac{-1}{\cos(\pi/4)} = \boxed{-\sqrt{2}}\end{aligned}$$

(xi) (10 pts)

First, factor the denominator: $x^2 + x - 2 = (x - 1)(x + 2)$.

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{\sin(x - 1)}{x^2 + x - 2} &= \lim_{x \rightarrow 1} \frac{\sin(x - 1)}{(x - 1)(x + 2)} \\ &= \left(\lim_{x \rightarrow 1} \frac{\sin(x - 1)}{x - 1} \right) \cdot \left(\lim_{x \rightarrow 1} \frac{1}{x + 2} \right)\end{aligned}$$

Let $u = x - 1$. As $x \rightarrow 1$, $u \rightarrow 0$.

$$\left(\lim_{u \rightarrow 0} \frac{\sin u}{u} \right) \cdot \left(\frac{1}{1 + 2} \right) = (1) \cdot \left(\frac{1}{3} \right) = \boxed{\frac{1}{3}}$$

(xii) (10 pts)

We are evaluating the limit as x approaches 0 from the positive side.

$$\lim_{x \rightarrow 0^+} \frac{\log x}{x} = \lim_{x \rightarrow 0^+} \left(\frac{1}{x} \cdot \log x \right)$$

As $x \rightarrow 0^+$, $\frac{1}{x} \rightarrow +\infty$ and $\log x \rightarrow -\infty$.

$$\lim_{x \rightarrow 0^+} \frac{\log x}{x} = \boxed{-\infty}$$

(xiii) (10 pts)

This limit is of the form ∞^0 . Let $L = \lim_{x \rightarrow \infty} (e^x + x)^{1/x}$. We take the natural logarithm.

$$\ln L = \lim_{x \rightarrow \infty} \frac{\ln(e^x + x)}{x}$$

This is of the form $\frac{\infty}{\infty}$, so we can apply L'Hôpital's Rule.

$$\ln L = \lim_{x \rightarrow \infty} \frac{\frac{e^x + 1}{e^x + x}}{1} = \lim_{x \rightarrow \infty} \frac{e^x + 1}{e^x + x} = \lim_{x \rightarrow \infty} \frac{1 + e^{-x}}{1 + \frac{x}{e^x}} = \frac{1 + 0}{1 + 0} = 1$$

Since $\ln L = 1$, we have $L = \boxed{e}$.

(xiv) (10 pts)

This limit is of the form 1^∞ . Let $L = \lim_{x \rightarrow 0^+} (\cos x)^{1/x^2}$. We take the natural logarithm.

$$\ln L = \lim_{x \rightarrow 0^+} \frac{\ln(\cos x)}{x^2}$$

This is of the form $\frac{0}{0}$. Applying L'Hôpital's Rule twice:

$$\ln L = \lim_{x \rightarrow 0^+} \frac{\frac{-\sin x}{\cos x}}{2x} = \lim_{x \rightarrow 0^+} \frac{-\tan x}{2x} = \lim_{x \rightarrow 0^+} \frac{-\sec^2 x}{2} = -\frac{1}{2}$$

Since $\ln L = -1/2$, the limit is $L = \boxed{e^{-1/2}}$ or $\boxed{\frac{1}{\sqrt{e}}}$.

Exercise 3.2 (30 pts)

Answer Summary

- (i) 1
 - (ii) 0
 - (iii) See the graph below.
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(i) (10 pts)

To evaluate $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$, let $t = \frac{1}{x}$. As $x \rightarrow \infty$, $t \rightarrow 0^+$.

$$\lim_{x \rightarrow \infty} x \sin \frac{1}{x} = \lim_{t \rightarrow 0^+} \frac{1}{t} \sin t = \lim_{t \rightarrow 0^+} \frac{\sin t}{t} = \boxed{1}$$

(ii) (10 pts)

To evaluate $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$, we use the Squeeze Theorem.

$$-1 \leq \sin \frac{1}{x} \leq 1$$

Multiplying by $|x|$ (for $x \neq 0$):

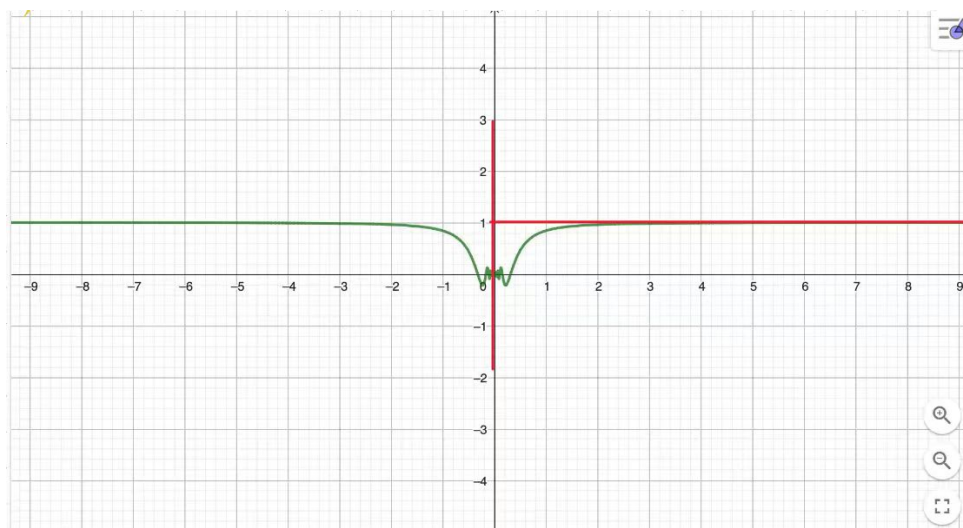
$$-|x| \leq x \sin \frac{1}{x} \leq |x|$$

Since $\lim_{x \rightarrow 0} (-|x|) = 0$ and $\lim_{x \rightarrow 0} |x| = 0$, by the Squeeze Theorem:

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = \boxed{0}$$

(iii) (10 pts)

The graph of $y = x \sin(1/x)$ illustrates the results. As $x \rightarrow \pm\infty$, the graph approaches the horizontal asymptote $y = 1$. As $x \rightarrow 0$, the function oscillates between $y = x$ and $y = -x$, converging to 0.



Exercise 3.3 (10 pts)

Given $|f(x)| \leq x^2$ for all $x \in \mathbb{R}$.

To calculate $f(0)$: Substitute $x = 0$ into the inequality: $|f(0)| \leq 0^2 \implies |f(0)| \leq 0$. Since absolute value cannot be negative, $|f(0)| = 0$, which implies $\boxed{f(0) = 0}$.

To calculate $f'(0)$: By the definition of the derivative: $f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$. From the given inequality, $|f(h)| \leq h^2$. For $h \neq 0$, divide by $|h|$ to get $\left| \frac{f(h)}{h} \right| \leq |h|$. This is equivalent to $-|h| \leq \frac{f(h)}{h} \leq |h|$. By the Squeeze Theorem, since $\lim_{h \rightarrow 0} (\pm|h|) = 0$, we conclude $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$. Therefore, $\boxed{f'(0) = 0}$.

Exercise 3.4 (10 pts)

Given f is an odd function and differentiable everywhere. We want to show that for any $b > 0$, there exists $c \in (-b, b)$ such that $f'(c) = f(b)/b$.

We apply the Mean Value Theorem (MVT) to f on the interval $[-b, b]$. 1. ****Continuity****: Since f is differentiable everywhere, it is continuous on $[-b, b]$. 2. ****Differentiability****: f is differentiable on $(-b, b)$. By the MVT, there exists a $c \in (-b, b)$ such that:

$$f'(c) = \frac{f(b) - f(-b)}{b - (-b)}$$

Since f is an odd function, $f(-b) = -f(b)$. Substituting this in:

$$f'(c) = \frac{f(b) - (-f(b))}{2b} = \frac{2f(b)}{2b}$$

Thus, we have shown that $\boxed{f'(c) = \frac{f(b)}{b}}$.

Exercise 3.5 (10 pts)

Evaluate $\lim_{x \rightarrow \infty} [x - x^2 \ln(\frac{1+x}{x})]$. This is an $\infty - \infty$ form. Let $t = 1/x$. As $x \rightarrow \infty$, $t \rightarrow 0^+$.

$$\lim_{t \rightarrow 0^+} \left[\frac{1}{t} - \frac{1}{t^2} \ln(1+t) \right] = \lim_{t \rightarrow 0^+} \frac{t - \ln(1+t)}{t^2}$$

This is a $\frac{0}{0}$ form, so we use L'Hôpital's Rule.

$$\lim_{t \rightarrow 0^+} \frac{1 - \frac{1}{1+t}}{2t} = \lim_{t \rightarrow 0^+} \frac{\frac{t}{1+t}}{2t} = \lim_{t \rightarrow 0^+} \frac{1}{2(1+t)} = \frac{1}{2(1+0)} = \boxed{\frac{1}{2}}$$

Exercise 3.6 (20 pts)

Given $f(x) = \begin{cases} |x|^x, & x \neq 0 \\ 1, & x = 0 \end{cases}$.

Answer Summary

- (i) The function f is continuous at 0.
 - (ii) $f'(0)$ does not exist.
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(i) (10 pts) Show that f is continuous at 0.

We must show that $\lim_{x \rightarrow 0} f(x) = f(0)$. We are given $f(0) = 1$. We evaluate $\lim_{x \rightarrow 0} |x|^x$. This is a 0^0 form. Let $L = \lim_{x \rightarrow 0} |x|^x$.

$$\ln L = \lim_{x \rightarrow 0} x \ln |x| = \lim_{x \rightarrow 0} \frac{\ln |x|}{1/x}$$

This is a $\frac{-\infty}{\infty}$ form. By L'Hôpital's Rule:

$$\ln L = \lim_{x \rightarrow 0} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0} (-x) = 0$$

Since $\ln L = 0$, $L = e^0 = 1$. Since $\lim_{x \rightarrow 0} f(x) = 1 = f(0)$, the function f is continuous at 0.

(ii) (10 pts) Calculate $f'(0)$.

By the definition of the derivative: $f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h|^h - 1}{h}$. This is a $\frac{0}{0}$ form. Consider the right-hand limit ($h \rightarrow 0^+$) and apply L'Hôpital's Rule.

$$\lim_{h \rightarrow 0^+} \frac{h^h - 1}{h} = \lim_{h \rightarrow 0^+} \frac{\frac{d}{dh}(h^h)}{1} = \lim_{h \rightarrow 0^+} h^h (\ln h + 1)$$

We know $\lim_{h \rightarrow 0^+} h^h = 1$ and $\lim_{h \rightarrow 0^+} \ln h = -\infty$. Thus, the right-hand limit is $1 \cdot (-\infty + 1) = -\infty$. Since the limit is not a finite number, the derivative at $x = 0$ does not exist. Therefore, $f'(0)$ does not exist.

Exercise 3.7 (10 pts)

Answer Summary

- For $x \neq 0$, $f'(x) = nx^{n-1} \sin(1/x) - x^{n-2} \cos(1/x)$.
- For $x = 0$:
 - if $n = 1$, $f'(0)$ does not exist.
 - if $n \geq 2$, $f'(0) = 0$.

Case 1: $x \neq 0$ Using the product and chain rules:

$$f'(x) = \frac{d}{dx}(x^n \sin(1/x)) = \boxed{nx^{n-1} \sin(1/x) - x^{n-2} \cos(1/x)}$$

Case 2: $x = 0$ Using the limit definition of the derivative:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^n \sin(1/h)}{h} = \lim_{h \rightarrow 0} h^{n-1} \sin(1/h)$$

The value of this limit depends on n .

- **If $n = 1$:** The limit becomes $\lim_{h \rightarrow 0} \sin(1/h)$. This limit does not exist because $\sin(1/h)$ oscillates between -1 and 1 as h approaches 0. Thus, for $n = 1$, $\boxed{f'(0) \text{ does not exist}}$.
- **If $n > 1$ (i.e., $n \geq 2$ since $n \in \mathbb{N}$):** The exponent $n-1$ is at least 1. By the Squeeze Theorem, since $-1 \leq \sin(1/h) \leq 1$, we have $-|h^{n-1}| \leq h^{n-1} \sin(1/h) \leq |h^{n-1}|$. As $h \rightarrow 0$, both $-|h^{n-1}|$ and $|h^{n-1}|$ approach 0. Thus, for $n \geq 2$, $\boxed{f'(0) = 0}$.